

Basic Electronics

Tutorial 1

1. The total charge flowing out of one end of a small copper wire and into an unknown device is determined to follow the relationship $q(t) = 5e^{-t/2}$ C, where t is expressed in seconds. Calculate the current flowing into the device, taking note of the sign. Draw the waveforms of $q(t)$ and $i(t)$ as a function of time.
2. A new type of device appears to accumulate charge according to the expression $q(t) = 10t^2 - 22t$ mC (t in s). (a) In the interval $0 \leq t < 5$ s, at what time does the current flowing into the device equal zero? (b) Sketch $q(t)$ and $i(t)$ over the interval $0 \leq t < 5$ s.
3. The current waveform depicted below Fig. 1 is characterized by a period of 4 s. (a) What is the average value of the current over a single period? (b) Compute the average current over the interval $1 < t < 3$ s. (c) If $q(0) = 1$ C, sketch $q(t)$, $0 < t < 4$ s.

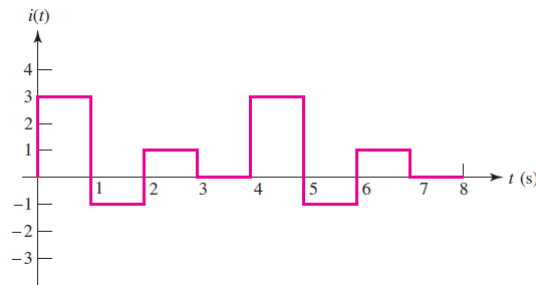


Figure 1: Prob.3

4. Determine the power absorbed by each of the elements in Fig. 2.

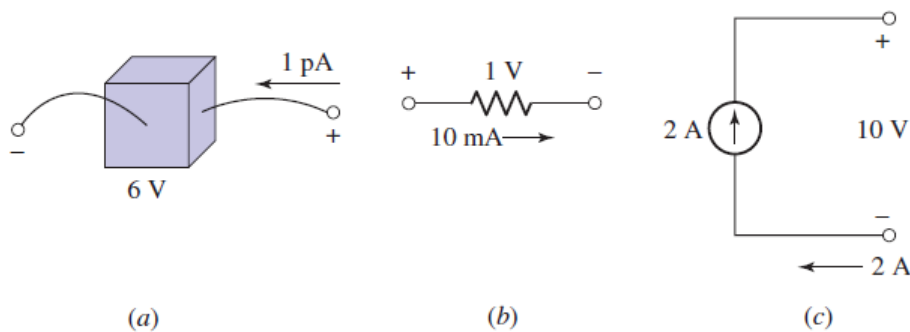


Figure 2: Prob.4

5. Determine the power supplied by the leftmost element in the circuit of Fig. 3.
6. The current-voltage characteristic of a silicon solar cell exposed to direct sunlight at noon in Florida during midsummer is given in Fig. 4. It is obtained by placing different-sized resistors across the two terminals

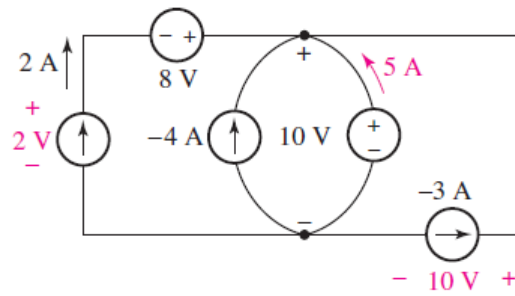


Figure 3: Prob.5

of the device and measuring the resulting currents and voltages. (a) What is the value of the short-circuit current? (b) What is the value of the voltage at open circuit? (c) Estimate the maximum power that can be obtained from the device.

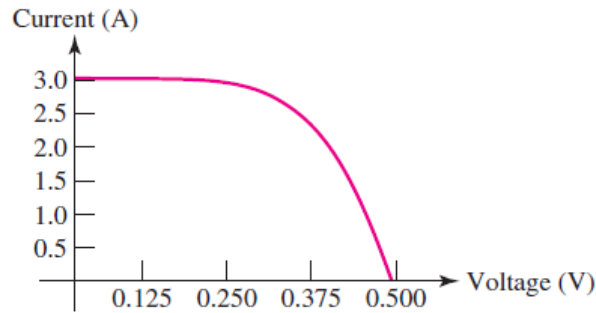


Figure 4: Prob.6

7. Refer to the circuit represented in Fig. 5, while noting that the same current flows through each element. The voltage-controlled dependent source provides a current which is 5 times as large as the voltage V_x . (a) For $V_R = 10$ V and $V_x = 2$ V, determine the power absorbed by each element. (b) Is element A likely a passive or active source? Explain.

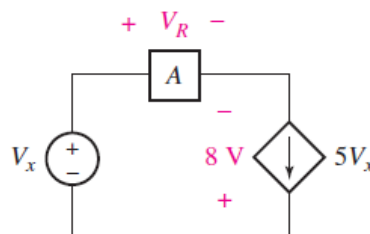


Figure 5: Prob.7

8. The dependent source in the circuit of Fig. 6 provides a voltage whose value depends on the current i_x . What value of i_x is required for the dependent source to be supplying 1 W?

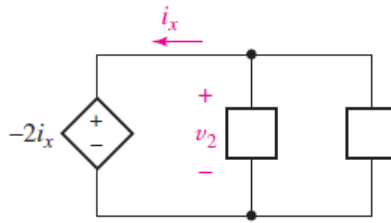


Figure 6: Prob.8

9. Real resistors can only be manufactured to a specific tolerance, so that in effect the value of the resistance is uncertain. For example, a $1\ \Omega$ resistor specified as 5% tolerance could in practice be found to have a value anywhere in the range of 0.95 to $1.05\ \Omega$. Calculate the voltage across a $2.2\ \text{k}\Omega$ 10% tolerance resistor if the current flowing through the element is (a) $1\ \text{mA}$; (b) $4 \sin 44t\ \text{mA}$.
10. For each of the circuits in Fig. 7, find the current I and compute the power absorbed by the resistor.

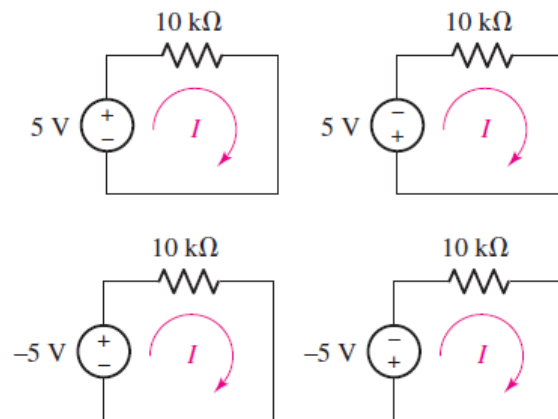


Figure 7: Prob.10

11. Sketch the power absorbed by a $100\ \Omega$ resistor as a function of voltage over the range $-2\text{V} \leq V_{\text{resistor}} \leq +2\text{V}$.